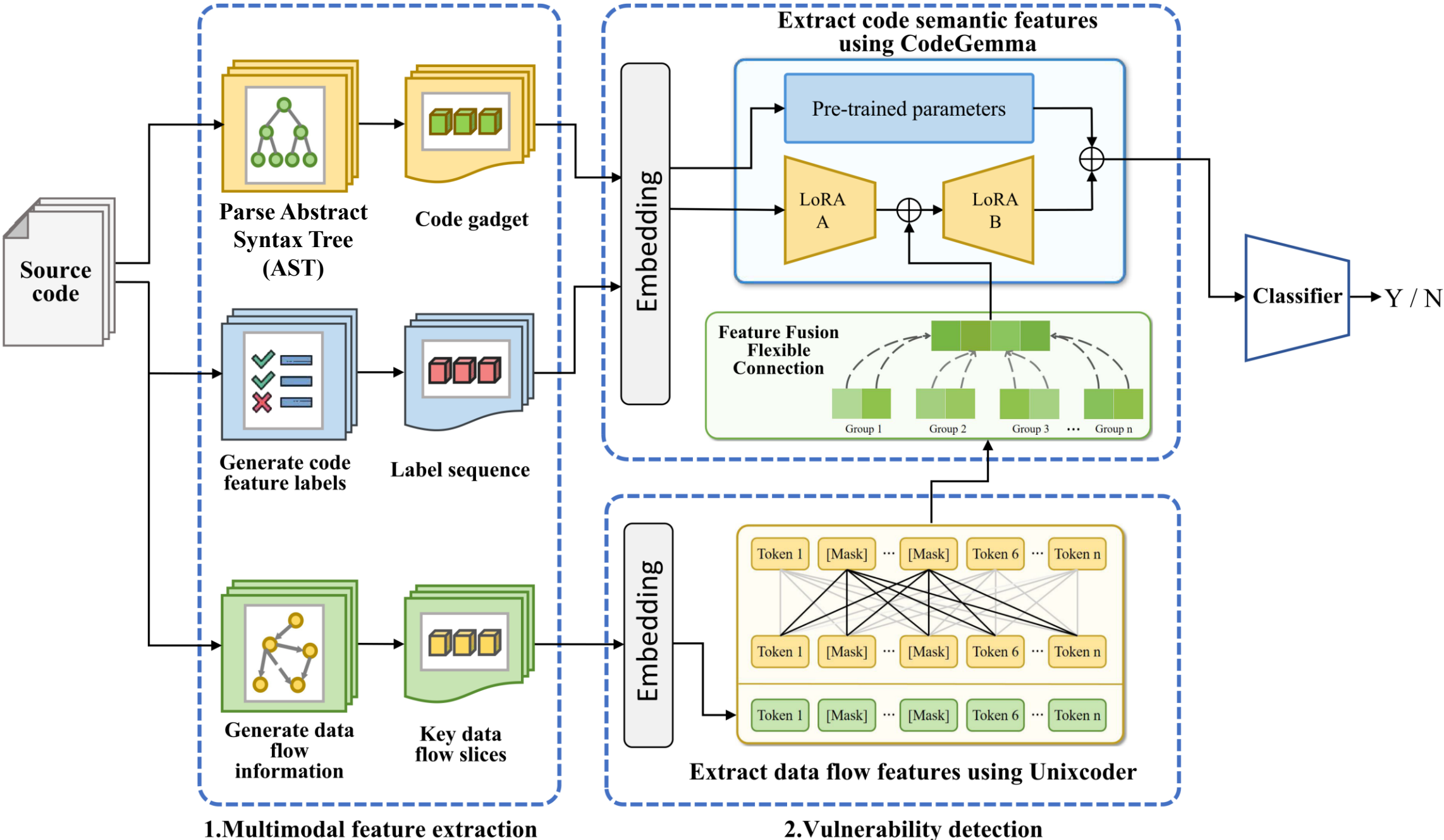
Source code
Parse Abstract Syntax Tree (AST)
Code gadget
Generate code feature labels
Label sequence
Generate data flow information
Key data flow slices
1.Multimodal feature extraction
Extract code semantic features using CodeGemma
Embedding
Pre-trained parameters
LoRA A
LoRA B
Feature Fusion Flexible Connection
Group 1
Group 2
Group 3
Group n
Classifier
Y / N
Embedding
Token 1
[Mask]
Token 6
Token n
Extract data flow features using Unixcoder
2.Vulnerability detection